NAME OF CANDIDATE: .....
STUDENT NUMBER: .....

THE UNIVERSITY OF NEW SOUTH WALES

SCHOOL OF MATHEMATICS AND STATISTICS

Sample Mid-session Test

MATH3871/MATH5960

Bayesian Inference and Computation

(1) TIME ALLOWED – 1.5 hours

(2) TOTAL NUMBER OF QUESTIONS – 4

(3) ANSWER ALL QUESTIONS

(4) THE QUESTIONS ARE **NOT** OF EQUAL VALUE

(5) THIS PAPER MAY **NOT** BE RETAINED BY THE CANDIDATE

(6) ONLY CALCULATORS WITH AN AFFIXED "UNSW APPROVED" STICKER MAY BE USED.

All answers must be written in ink. Except where they are expressly required pencils may only be used for drawing, sketching or graphical work.

1. [**11 marks**] Consider the problem of expressing subjective prior beliefs about parameters values via prior distributions in Bayesian statistical inference. Give **brief** descriptions in words for the following questions:

a) [**3 marks**] What is a conjugate prior? Why do we use conjugate priors and when is it not appropriate to use them?

b) [**3 marks**] What is an improper prior? Give an example of one and state the conditions under which an improper prior can be used for inference.

c) [**2 marks**] What is Jeffreys' prior and when is it used?

d) [**3 marks**] Consider a (general) mixture prior distribution for θ of the form

$$\pi(\theta) = \sum_{i=1}^{k} w_i \pi_i(\theta) \qquad \sum_{i=1}^{k} w_i = 1.$$

Give an example of when the mixture prior distribution may become an appropriate expression for the prior belief. What is the role of the weights w_i? Do the distributions $\pi_i(\theta)$ have to take the same form?

2. [**20 marks**] Consider a vector $\mathbf{y} = (y_1, \ldots, y_n)^\top$ of n iid observations from the distribution $N(\mu, \sigma^2)$, where mean μ and variance σ^2 are unknown. Then each y_i has probability density function of the form

$$f(y_i) = \frac{1}{\sqrt{2\pi}\sigma} \exp(-\frac{1}{2\sigma^2}(y_i - \mu)^2)$$

Suppose we take the vague prior density for μ and σ given by

$$\pi(\mu, \sigma^2) \propto (\sigma^2)^{-1}$$

and let

$$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (y_i - \bar{y})^2.$$

a) [**5 marks**] Derive the joint posterior density $\pi(\mu, \sigma^2|\mathbf{y})$, show that it is proportional to

$$\sigma^{-n-2} \exp\left(-\frac{1}{2\sigma^2}[(n-1)s^2 + n(\bar{y} - \mu)^2]\right).$$

b) [**3 marks**] Show that the conditional distribution $\pi(\mu|\sigma^2, \mathbf{y})$ is of the form

$$\mu|\sigma^2, \mathbf{y} \sim N(\bar{y}, \sigma^2/n).$$

c) [**5 marks**] Show that the marginal posterior distribution σ^2 is

$$\pi(\sigma^2|\mathbf{y}) \propto (\sigma^2)^{-(n+1)/2} \exp\left(-\frac{(n-1)s^2}{2\sigma^2}\right).$$

Please see over ...

d) **[7 marks]** Show that the marginal posterior distribution for μ is

$$\pi(\mu|\mathbf{y}) \propto \left[1 + \frac{n(\bar{y}-\mu)^2}{(n-1)s^2}\right]^{-n/2}.$$

Hint: You might like to make the substitutions

$$z = \frac{A}{2\sigma^2}, \quad A = (n-1)s^2 + n(\bar{y}-\mu)^2$$

and observe that the normalised Gamma distribution has the form

$$f(\theta) = \frac{\beta^\alpha}{\Gamma(\alpha)}\theta^{\alpha-1}e^{-\beta\theta}, \theta > 0.$$

3. **[10 marks]** In Lectures, you were given the rejection algorithm, which obtains a sample x from the distribution $f(x)$ via the following three steps:

1 Simulate x from $g(x)$

2 Simulate y from $U(0, Kg(x))$

3 Accept x if $y \leq f(x)$

a) **[2 marks]** The function $g(x)$ can be chosen arbitrarily, however in practice, a good choices of g is essential to the performance of rejection sampling. Give two features of $g(x)$ that are either essential or desirable for the algorithm.

b) **[2 marks]** Why is K needed in the algorithm?

c) **[4 marks]** What is the probability of accepting a single point x in the rejection step? Hence derive the overall acceptance probability for all samples simulated from $g(x)$.

d) **[2 marks]** Hence, using the acceptance probability given in part d), give an alternative algorithm (to the above) for rejection sampling.

4. **[14 marks]** Consider the loss function $L(\theta, d)$ which quantifies the loss achieved when making decision $d \in \mathcal{D}$ given a fixed parameter value θ.

a) **[2 marks]** What is the role of the loss function in Bayesian inference? Write down an expression for the optimal decision, d^*, in this setting.

b) **[6 marks]** Show that adopting the quadratic loss $L(\theta, d) = (\theta - d)^2$ results in the optimal parameter estimate $d^* = E(\theta|x)$, the posterior mean.

c) **[6 marks]** Derive the optimal parameter estimate, d^*, when using the 0-1 loss function

$$L(\theta, d) = \begin{cases} 0 & \text{if } |\theta - d| \leq \epsilon \\ 1 & \text{if } |\theta - d| > \epsilon \end{cases}.$$